

INTRODUCTION TO CLOUD COMPUTING



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Work Integrated Learning Programmes

AGENDA – CLOUD COMPUTING

Definitions, characteristics, and motivation

Evolution and origins

Service models



LET'S DISCUSS THE BASICS!

When we plug a phone into an outlet, we are consuming electricity as a **utility**.

We don't own the power plant

We don't manage the grid

We just consume what we need and pay for what we use.



What is the technical term that comes to your mind for this analogy?

Cloud computing applies this exact **utility model** to IT resources.

LET'S DISCUSS THE BASICS!

Just as the electrical outlet **abstracts away** the complexity of the power plant and the grid, cloud computing provides different levels of **abstraction** for the IT resources.

We utilize the useful function without needing to manage the complex internals.



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LET'S DISCUSS THE BASICS!

How is the cloud made possible?

Virtualization (Partitioning One into Many): Technology that allows us to take a single, powerful physical server and partition it into multiple, isolated virtual machines (VMs). It's the art of slicing up one big resource into many smaller, usable ones.

Distributed Systems (Combining Many into One): The Principle used to connect thousands of independent computers and have them work together as a single, resilient, and coherent system. It's the art of combining many resources to build a service that is far more powerful and reliable than any single machine."

LET'S DISCUSS THE BASICS!

Cloud computing masterfully combines the virtualization and distributed system concepts.



Let's take a cloud like AWS.

virtualization - slice up the physical servers into EC2 instances for customers.

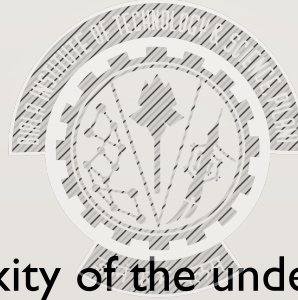
uses the principles of **distributed systems** to build hyper-scalable services like S3 or DynamoDB. They run across thousands of machines but act as a single, unified service for the end-user."

LET'S DISCUSS THE BASICS!

The best way to understand the business model of cloud computing is to think of it like the **electrical grid**—it transformed IT into a **utility**.

But how is this utility actually built?

It relies on two foundational technologies.



1) **Abstraction**, which hides the immense complexity of the underlying systems, just like an outlet hides the power plant.

2) Powered by the combination of the following:

Virtualization - partition massive servers into usable chunks

&

Distributed systems - weave thousands of these servers together into the powerful, resilient services we use every day.

.... LET'S DISCUSS THE BASICS!

Cluster, Grid and Cloud Computing – allow access to computing power in a fully virtualized manner

- Aggregation of resources
- Single System View
- Computing as a utility



- Utility Computing is a business model for on-demand delivery of computing power.
- Reason for the naming?

.... BASICS!

Cluster, Grid and Cloud Computing – allow access to computing power in a fully virtualized manner

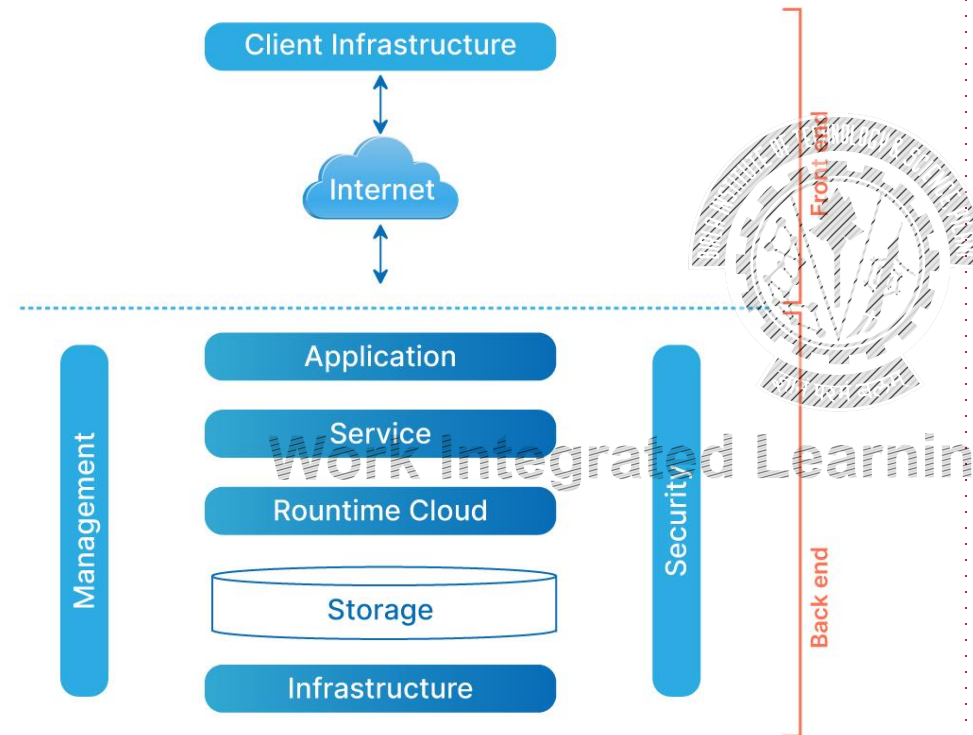
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CLOUD COMPUTING

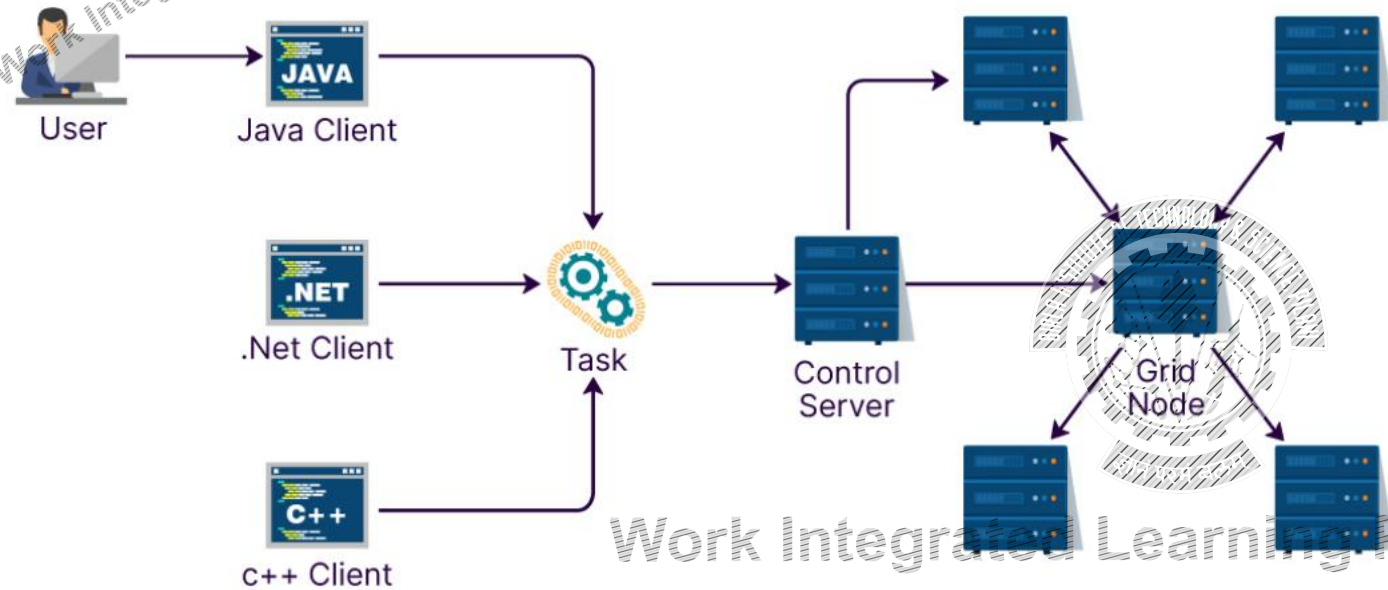
ARCHITECTURE OF CLOUD COMPUTING



- Service – 3 flavours
 - PaaS
 - SaaS
 - IaaS
- Runtime Cloud – VMs and Runtime environments

One of the ways of looking at cloud computing

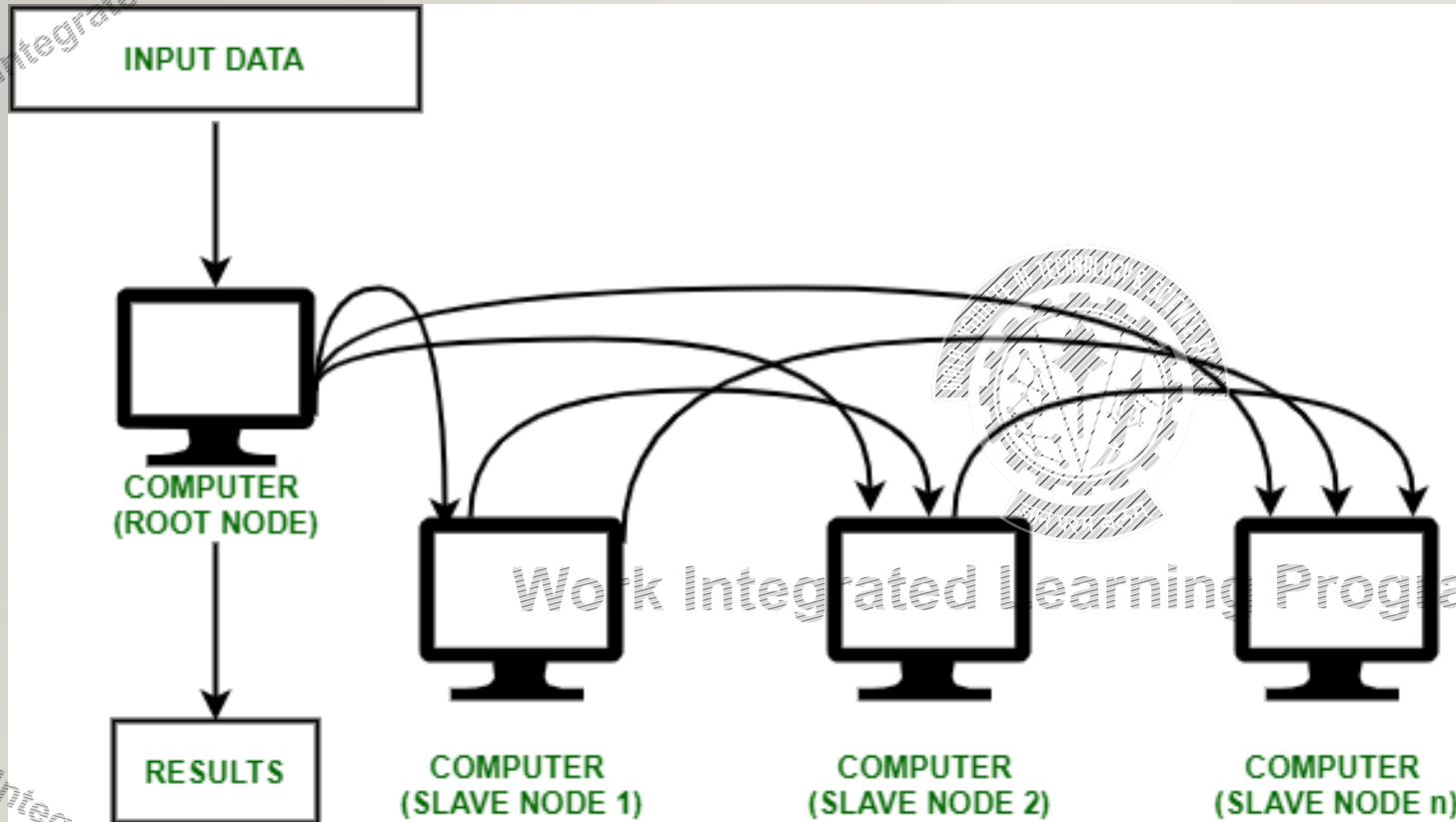
GRID COMPUTING



- Distributed Computing (Decentralized model)
- Solutions for Complex Problems
- Resources in the grid work together to complete a task
- Heterogeneous Nodes

Any complex and time consuming activity can be addressed

CLUSTER COMPUTING



- Homogeneous Nodes most of the times.
- Nodes mostly placed in the same location
- Whole System to be considered as one single system

To the outside world, it is just one system

CLUSTER COMPUTING – QUESTIONS TO CONSIDER



- Homogeneous Nodes most of the times – What can be the reason?
- Nodes mostly placed in the same location – What can be the reason?

CLUSTER COMPUTING – SALIENT POINTERS

- Homogeneous Nodes most of the times - Reason
 - ✓ **Challenging** when the nodes are not identical.
 - ✓ Heterogeneous nodes have different processing capabilities, operating systems and hardware. This would mean that we need specialized algorithms for effective task distribution and to maximize resource utilization.
 - ✓ So, a homogeneous setup is often preferred for **simplicity**.

Note: Heterogeneous clusters are common, especially in high-performance computing (HPC) and virtualized environments and they require specialized algorithms to effectively distribute tasks and maximize resource utilization.
- Nodes mostly placed in the same location – Reason
 - to minimize network latency
 - ensure high bandwidth for working together seamlessly (direct connections, physical proximity allows for the use of high speed dedicated network links)

INTERNET – THE ORIGIN



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INTERNET – THE ORIGIN

ARPANET (Advanced Research Projects Agency Network) was a pioneering computer network developed by the United States Department of Defense in 1969.

This is the definite **precursor** to the modern internet.

It enabled communication between research institutions and government agencies using innovative technologies like packet switching.

It was one of the first networks to implement the TCP/IP protocol suite - allow data to be broken into packets and transmitted across a decentralized network.

We can say that it all started technically with Arpanet

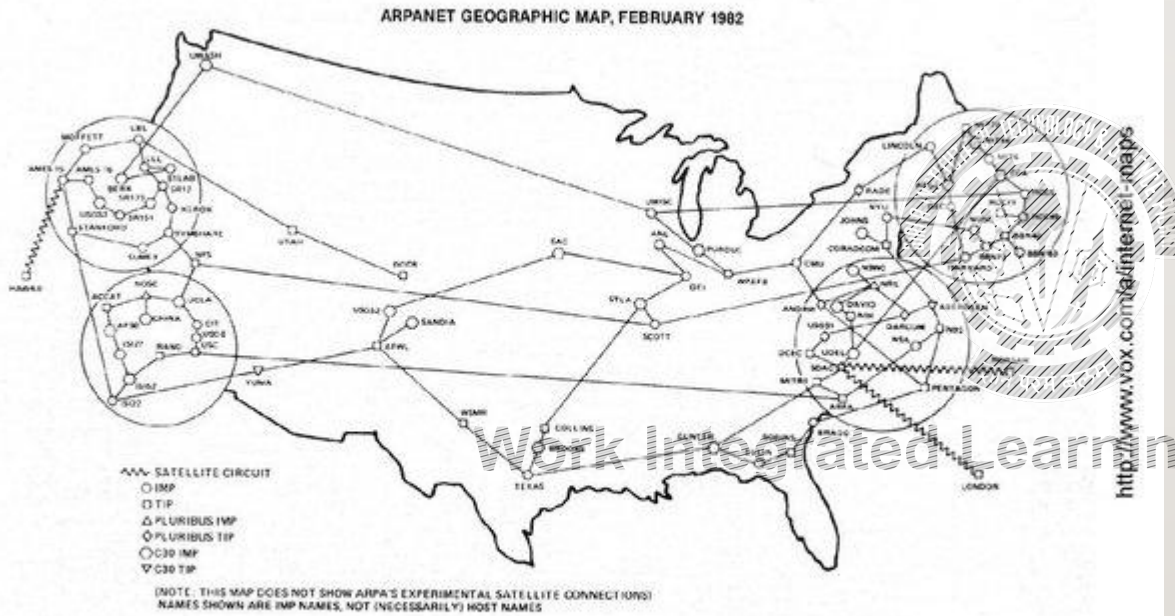
INTERNET – THE ORIGIN

- First Operation Packet Switching Network: Before ARPANET, networks were based on “Circuit Switching” like a telephone call where a dedicated connection is established for duration of the communication. Packet switching – Data is broken into small chunks (packets) and sent independently over a network. Packets are re-assembled at the destination.
- Packet Switching is the fundamental technology that makes the internet work
- This led to TCP/IP - Transmission Control Protocol/Internet Protocol. The protocol came based out of necessity as many similar networks that started to emerge after ARPANET had to communicate with each other. The initial protocol used by ARPANET was NCP (Network Control Protocol - the host to host protocol in a packet switched network. Packets would have a NCP with a destination header and message type information)

Arpanet gave a solid foundation base for the development of internet
Arpanet was built over ideas of its predecessors/research papers

WHY IS IT NAMED CLOUD?

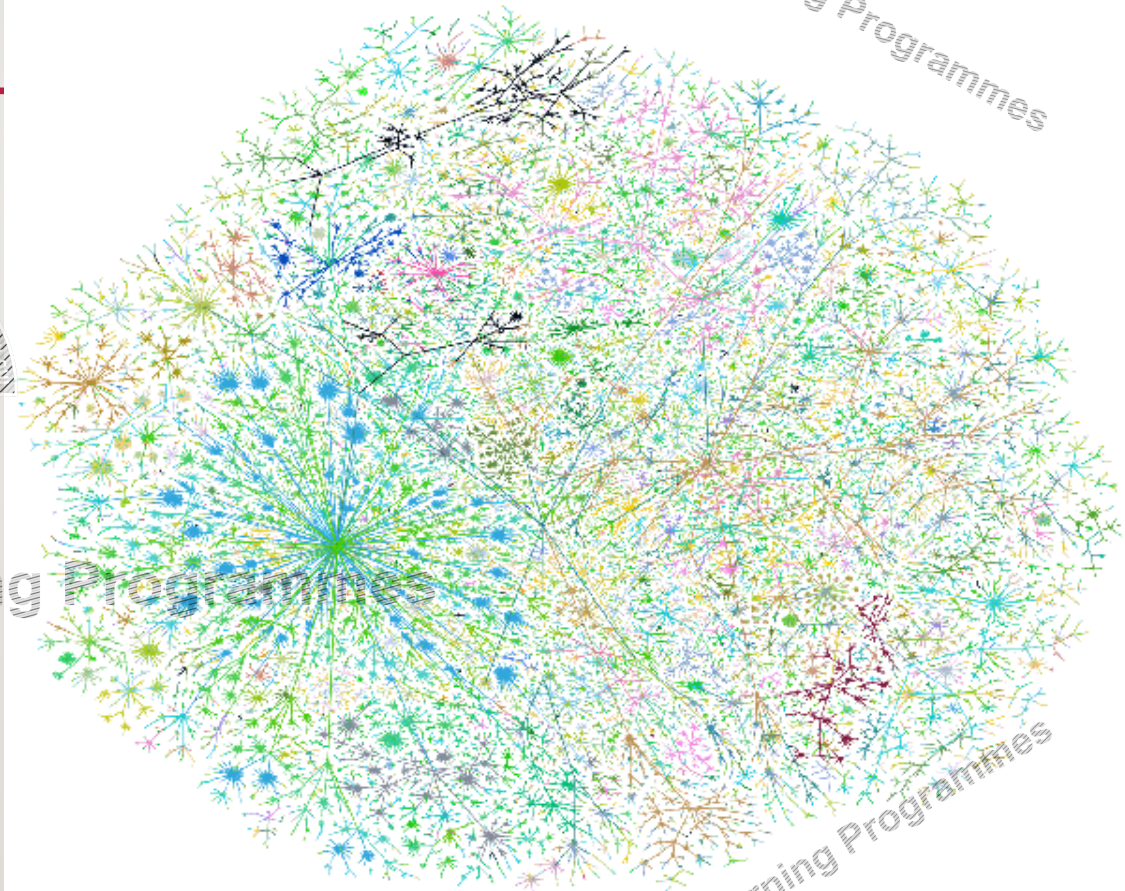
Logical Map of the ARPANet, February, 1982



By 1982, the network only had about 100 nodes. But that was enough to support an online community

10

1982



1998

WHY IS IT NAMED CLOUD?



Any Thought bubbles! ☺

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WHY IS IT NAMED CLOUD?



Inspiration from the cloud symbol!!!???

It could have as well been named anything else...
say Sky Computing!

Too many debates wrt who gets the credit for
coining the term

The origins of the term "cloud computing" are debatable.
There isn't one single person or company definitively credited with coining it.

WHY IS IT NAMED CLOUD?

Cloud has been used as a metaphor for a network.

Specifically for the internet this metaphor is the most widely accepted origin of the term. For decades, network diagrams used a cloud shape to represent the vast, interconnected network of servers, routers, and other devices.

The internal workings of these devices were not essential to the diagram's purpose and hence this abstracted representation of the internet became a common visual shorthand for the network itself.

When the concept of delivering computing resources and services over this network emerged, the metaphor stuck. **At some point in time, a specific term gets more traction and gets accepted**

CLOUD COMPUTING DEFINITIONS

Cloud computing has been coined as an umbrella term to describe a category of sophisticated on-demand computing services initially offered by commercial providers, such as Amazon, Google, and Microsoft. It denotes a model on which a computing infrastructure is viewed as a “cloud,” from which businesses and individuals access applications from anywhere in the world on demand. The main principle behind this model is offering computing, storage, and software “as a service.”

Cloud is a parallel and distributed computing system consisting of a collection of inter-connected and virtualized computers that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements (SLA) established through negotiation between the service provider and consumers.

Clouds are a large pool of easily usable and accessible virtualized resources (such as hardware, development platforms and/or services). These resources can be dynamically reconfigured to adjust to a variable load (scale), allowing also for an optimum resource utilization.

This pool of resources is typically exploited by a pay-per-use model in which guarantees are offered by the Infrastructure Provider by means of customized Service Level Agreements.”

So many definitions of Cloud Computing and the list continues

... CLOUD COMPUTING DEFINITIONS

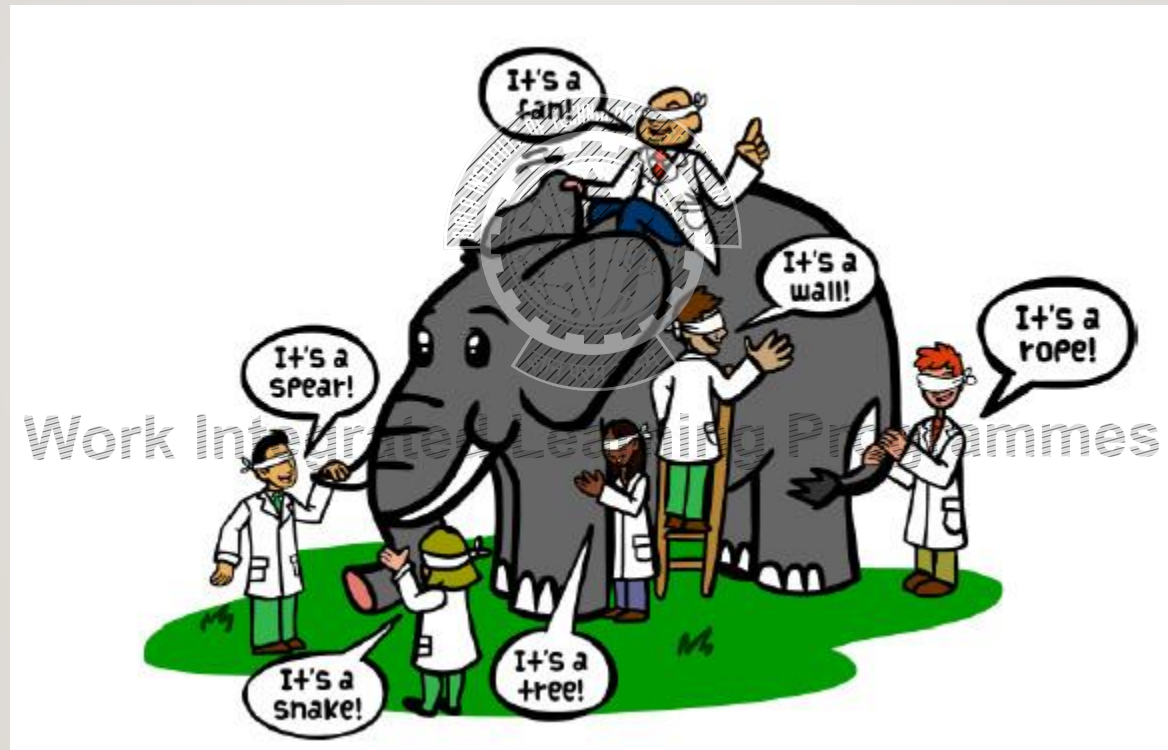
Clouds are hardware - based services offering compute, network, and storage capacity where: Hardware management is highly abstracted from the buyer, buyers incur infrastructure costs as variable OPEX, and infrastructure capacity is highly elastic.

A pay-per-use model for enabling available, convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Generic definition: Data center hardware and software that provide services.

So many definitions of Cloud Computing and the list continues

WHAT DO WE DO – WHICH ONE IS CORRECT



It all depends on the aspect from which we see/view

COMMON CHARACTERISTICS GIVEN BY NATIONAL INSTITUTE OF STANDARDS (NIST)

Pay-Per-Use (no ongoing commitment, utility prices) *

Elastic capacity (illusion of infinite resources)

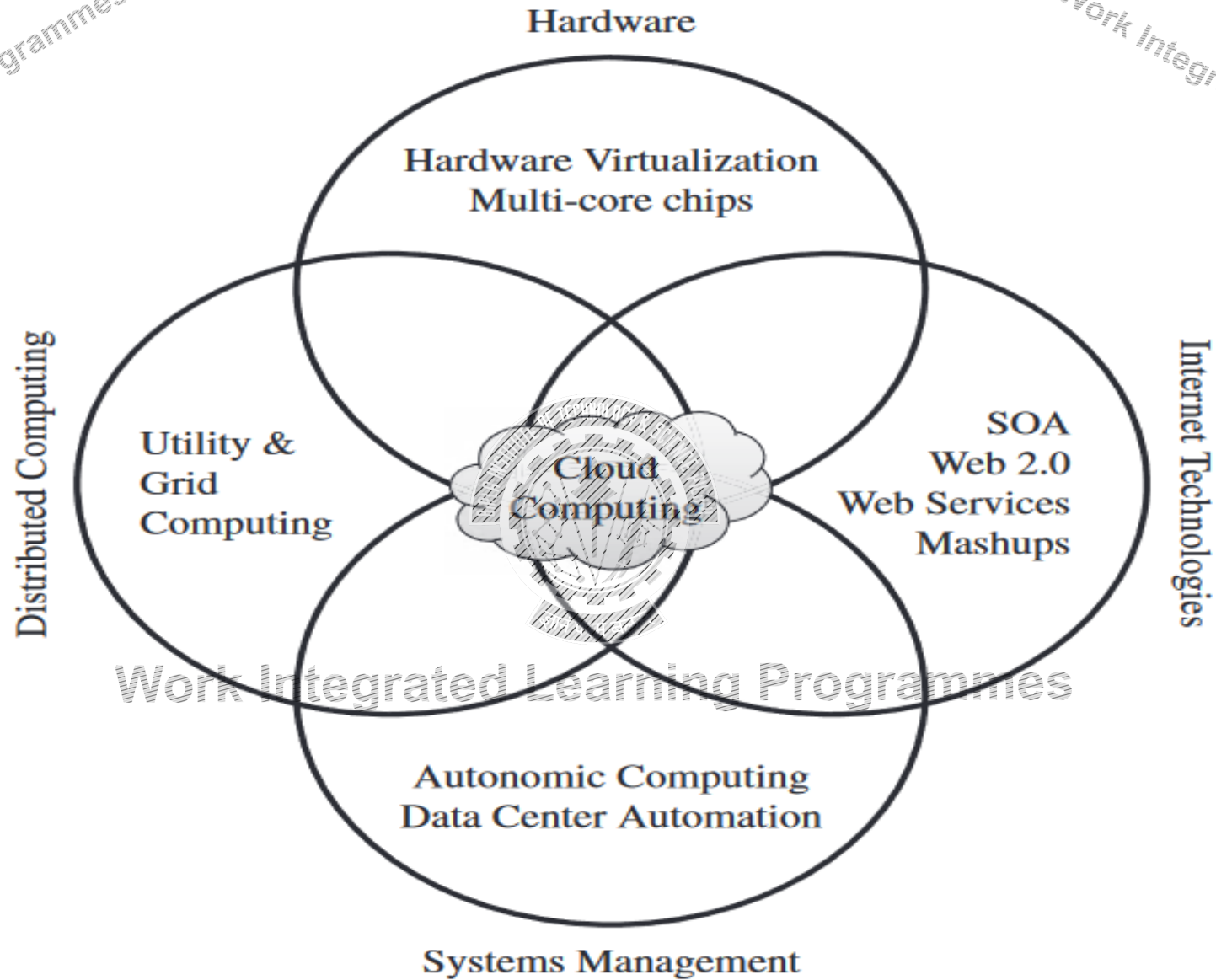
On Demand Self Service

Abstraction/Virtualisation of resources

Broad Network Access

* → Exceptions do happen as the concepts evolve For example – make a commitment and get discounts for usage. But, even in that case it is essentially Pay-Per-Use

Instead of getting drowned into the various perspectives, let us note down the common characteristics



Convergence of various advances enabled cloud computing

MOTIVATION

Let's find out by doing role plays

Proprietor of a Startup

A CEO of a well-established Company providing Software solutions

Head of a College

A Solution Architect



We saw the origin and various definitions. But, why Cloud?

KEYWORDS

Agility & Innovation

Cost

Scalability

OPEX / CAPEX

Ease of Management

Architecture Freedom

Infrastructure Freedom/Lower Cost of Ownership

Faster Return on Investment (ROI)



Well, there would have been so many positives that we noted during the roleplay

AREN'T THERE ANY ISSUES OR CONCERNS - IS IT ALL GREEN?

Security

Compliance

Interoperability and Vendor lock-in

I have no control!!!!

Very very minor: Connectivity issues (internet)



Each of these concerns (except the last!) do get addressed by Cloud Vendors and/or by the clients utilizing the Cloud

Every aspect has both positives and some negatives or at least concerns. Cloud Computing is not an exception

SERVICE MODELS - LAYERS AND TYPE OF CLOUD SERVICES

IaaS Infrastructure as a Service

PaaS Platform as a Service

SaaS Software as a Service



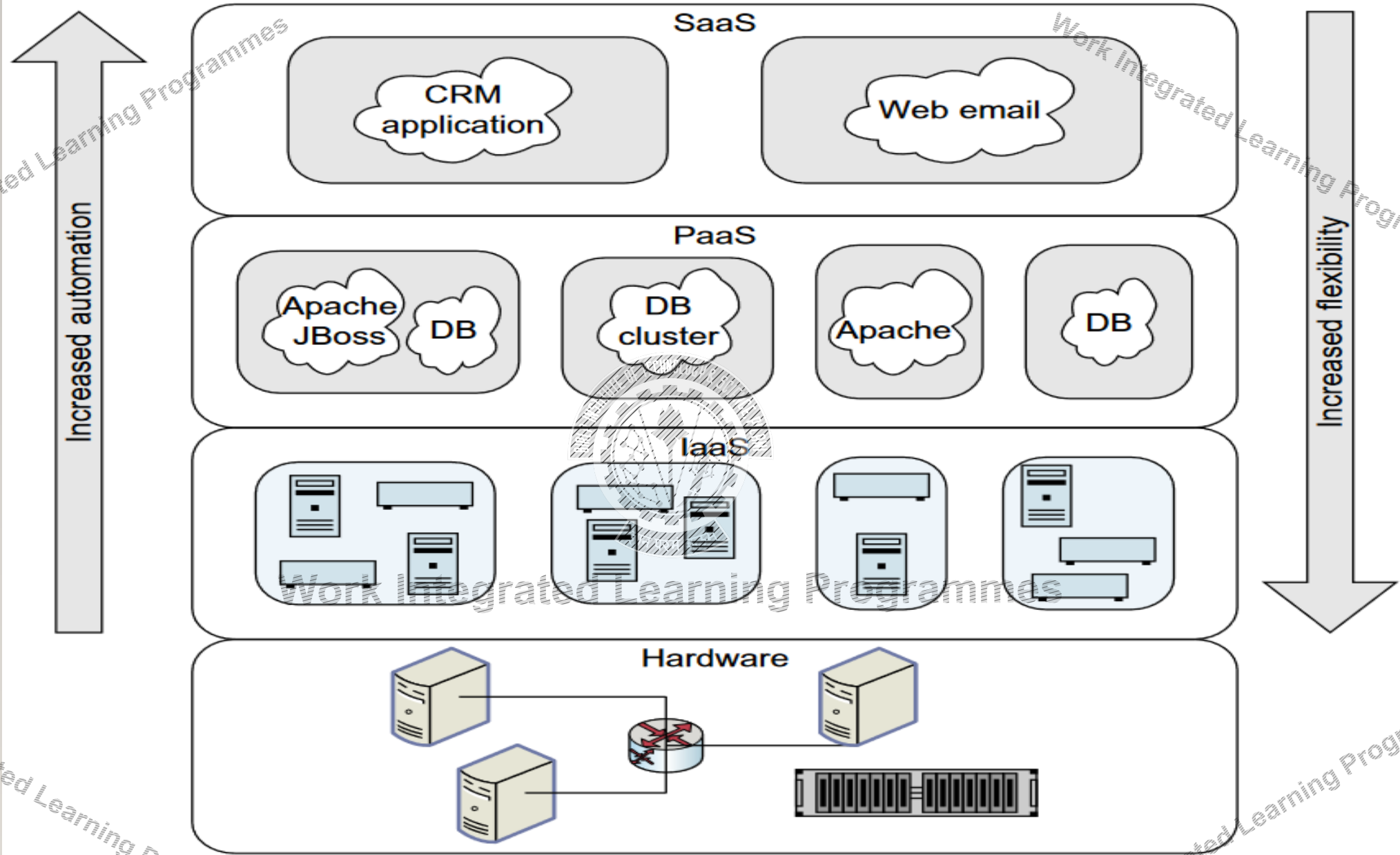
Focus is on a specific layer in a computer's runtime stack – the hardware, the system software (or platform) and the application, respectively

Abstraction level of the capability that is provided

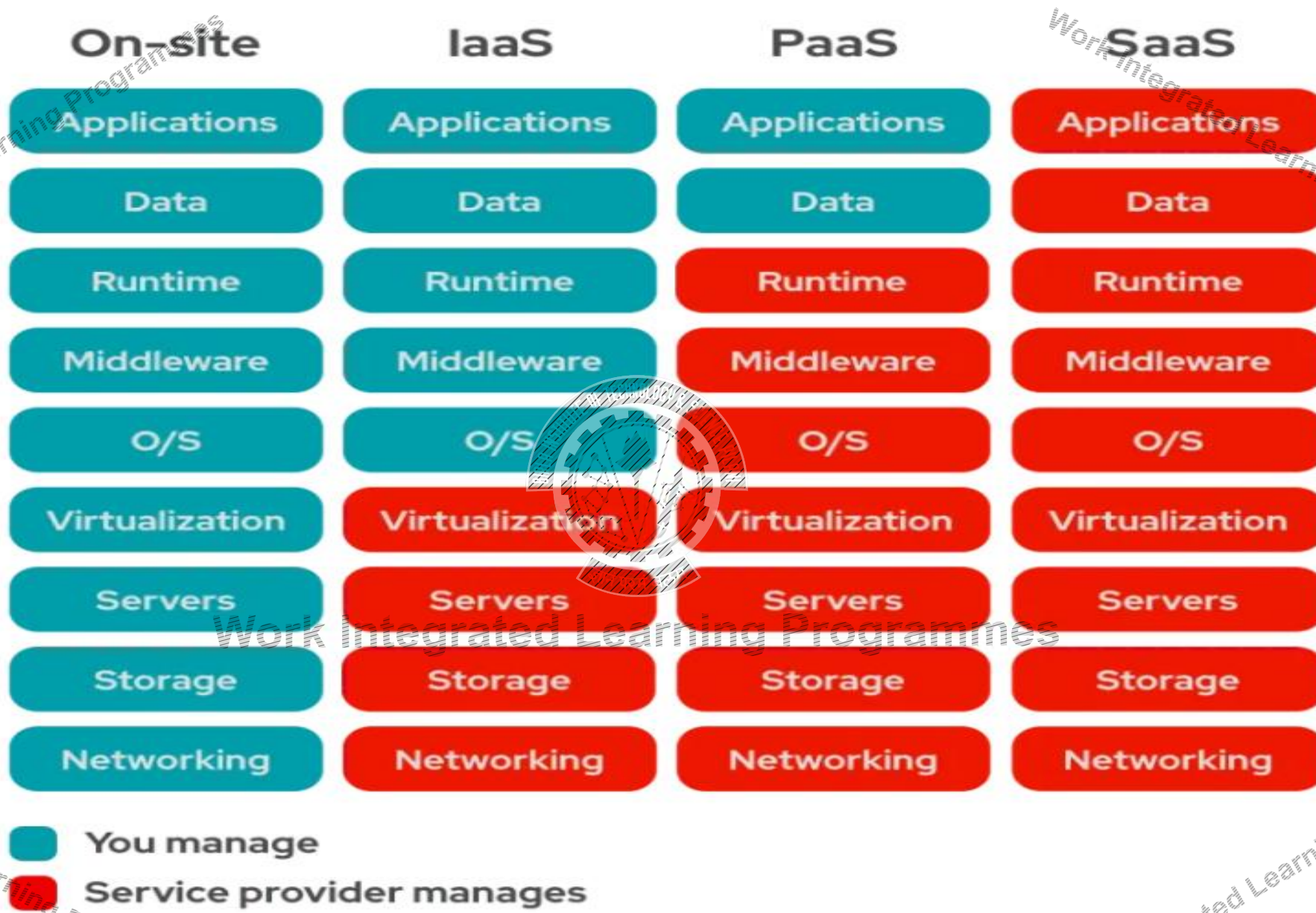
LAYERS AND TYPE OF CLOUD SERVICES

Service Class	Main Access & Management Tool	Service content
 SaaS	Web Browser	Cloud Applications Social networks, Office suites, CRM, Video processing
 PaaS	Cloud Development Environment	Cloud Platform Programming languages, Frameworks, Mashups editors, Structured data
 IaaS	Virtual Infrastructure Manager	Cloud Infrastructure Compute Servers, Data Storage, Firewall, Load Balancer

Services of a higher layer can be composed from the services of the underlying layer

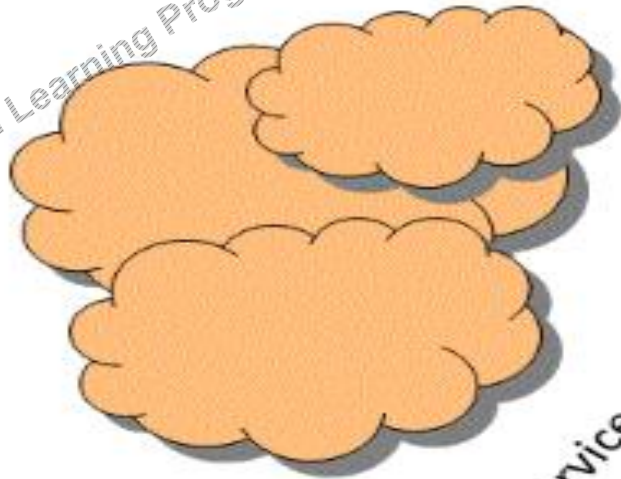


Flexibility vs Ease of Maintenance and Usage
TI: Moving to the Cloud



The question of what you do manage – summarizes it all!

Cloud Services



Applications
(SaaS)

Platform as a Service



Developers Services
(PaaS)

Physical Server; CPU, RAM
Storage
Data Center Networks
(IaaS)



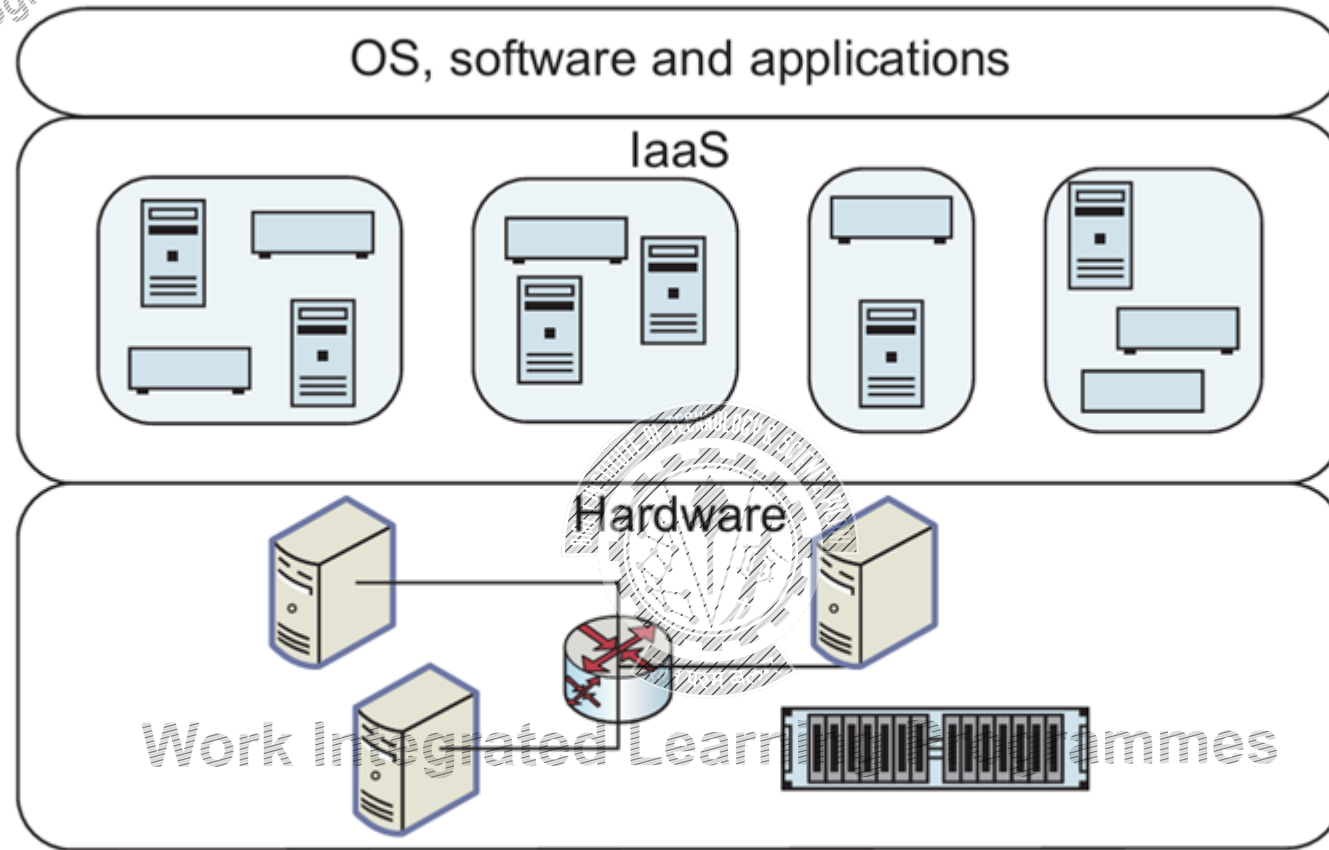
Infrastructure as a Service

IAAS – COMPUTE & STORAGE AS A SERVICE

NIST definition for IaaS: The capability provided to the consumer is to provision processing, storage, networks and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).

Ownership of the VM

User uses as if it is his/her own machine



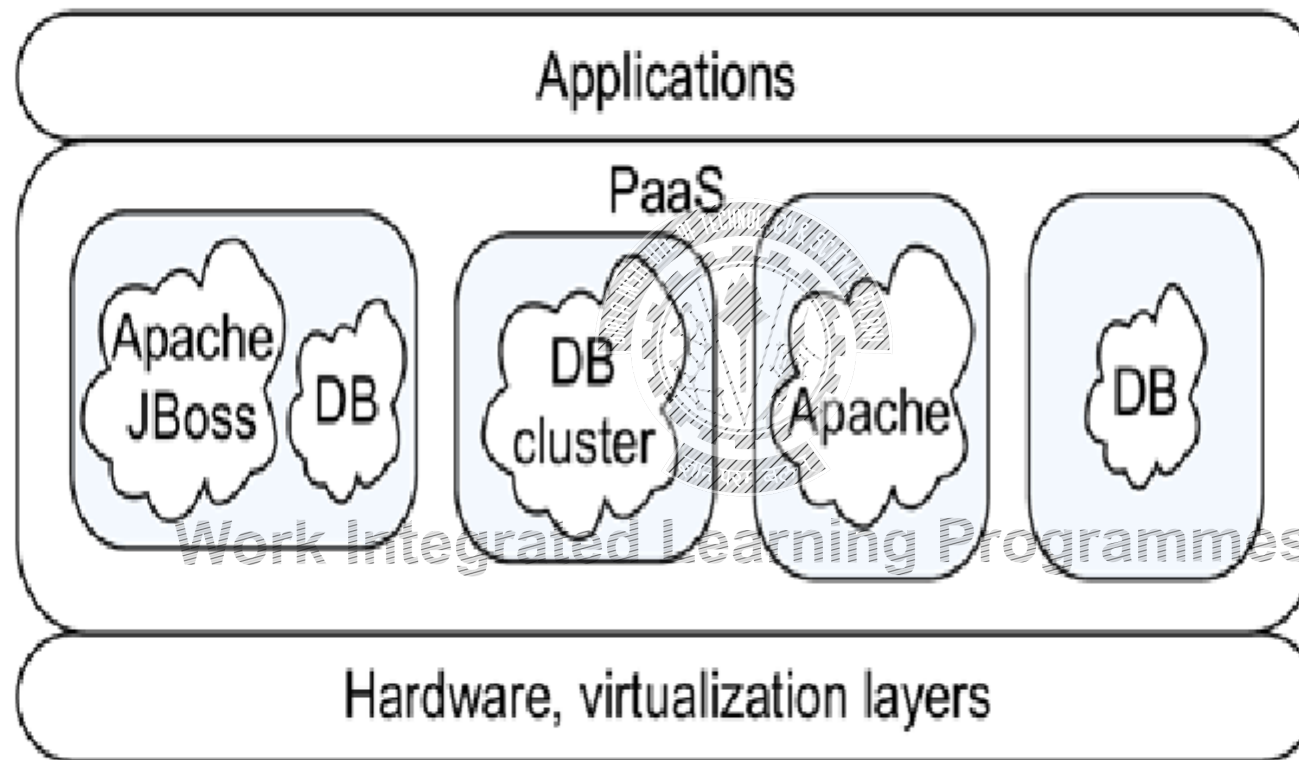
Questions: When we start, stop or terminate a VM, what happens to the physical hardware?
When aspects are considered by the cloud providers for the billing when a VM is started, stopped and terminated?
Would the billing aspects/considerations of the various cloud providers be the same?

PAAS – SYSTEM STACK/PLATFORM FOR APP DEPLOYMENT AS A SERVICE

NIST Definition for PaaS: The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.

Focus on your
application/development

User can focus on the development goal



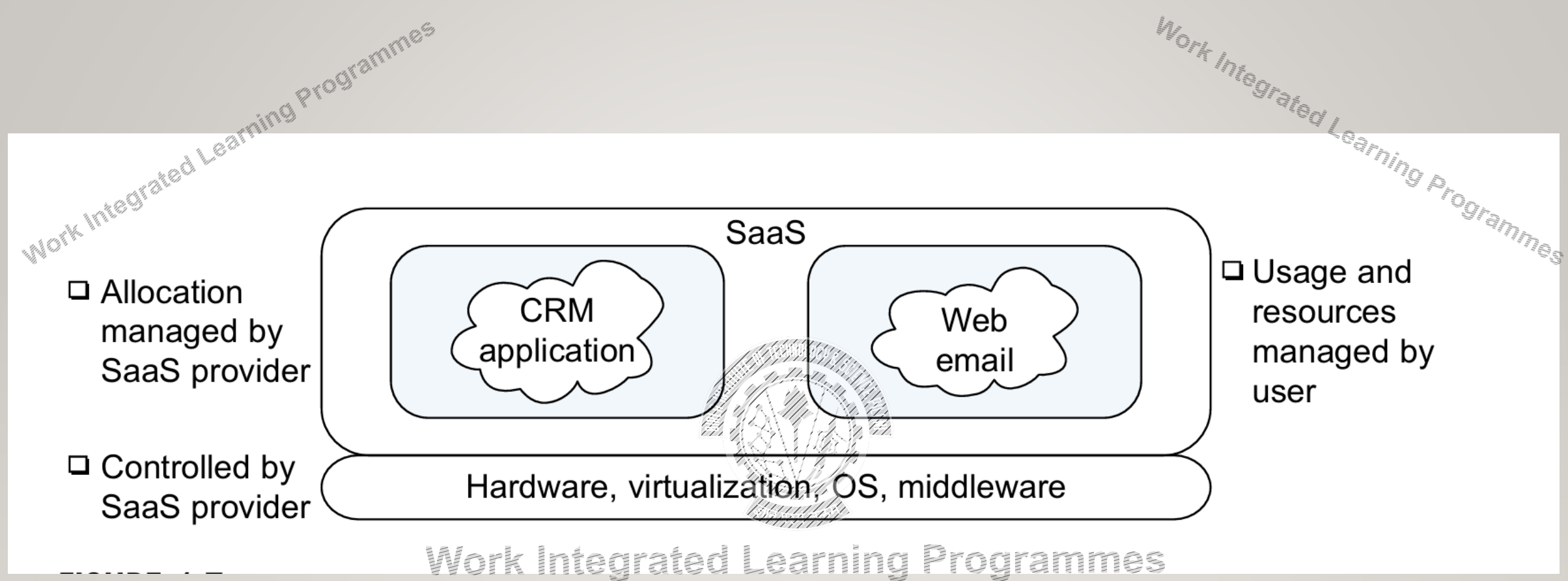
SAAS

The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email). The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

Need just a Web Browser!

Easiest one among the three

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IaaS

Amazon Web Services

Google Cloud

Microsoft Azure

IBM Cloud

PaaS

Google App Engine

Red Hat OpenShift

Heroku

Apprenda

SaaS

HubSpot

JIRA

Dropbox

DocuSign

Ease of Usage vs Ease of Control and Customization